Deepak Prasad S

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Summary

AI Engineering undergraduate specializing in **machine learning**, **LLM applications**, and **scalable backend APIs**. Experienced building production-grade **RAG pipelines**, **vector search systems**, and **NLP models**. Proficient in **Python**, **FastAPI**, and modern **GenAI** tooling. Seeking AI/ML engineering roles to apply skills in **natural language processing**, **deep learning**, and **intelligent system design**.

Technical Skills

Languages: Python, Java, C, SQL, HTML, CSS, JavaScript
Machine Learning / AI: Scikit-learn, Pandas, NumPy, Sentence-Transformers, NLP, Prompt Engineering
LLM & Generative AI: HuggingFace Transformers, RAG Architecture, Vector Databases, Groq API.
Backend & Databases: FastAPI, Flask MySQL, MongoDB, ChromaDB
Cloud & DevOps: Git, GitHub, Vercel, Aiven, Docker, CI/CD.

Experience

Associate AI Developer Intern *Aug 2026 – Present*
Expora Database Consultancy Services *Bengaluru, India (On-site)*

- Designed and built **OpenCV**-based computer vision systems tailored to client requirements.
- Developed **RAG-based applications** with **LLMs** and **vector search** for context-aware answers over client data.
- Worked directly with clients to gather requirements and deliver completed, end-to-end applications.

Project Experience

Real-Time Voice Intelligence Platform *Personal Project*
Python | FastAPI | React | WhisperX | PyTorch GitHub

- Architected real-time transcription system using **FastAPI WebSockets** and **faster-whisper + WhisperX** forced alignment, achieving **<1s** first-transcript latency, **95%+ word accuracy**, and **2× real-time** CPU processing speed across 99 languages.
- Engineered multi-speaker diarization pipeline with **Pyannote.audio 3.1**, Resemblyzer embeddings, and custom **Wav2Vec2** overlap classifier for speaker-attributed transcripts.
- Integrated **Groq LLaMA 3.3-70B** for AI meeting insights (summaries, action items, Q&A) with **React + TypeScript** frontend and async model loading for zero cold-start latency.

Production-Grade RAG Pipeline for Document Intelligence *Personal Project*
Python | FastAPI | ChromaDB | Sentence-Transformers | Vector Search GitHub

- Built end-to-end **RAG** system for PDF Q&A using **FastAPI**, **ChromaDB** vector storage, sentence embeddings, and metadata-filtered retrieval.
- Designed chunking + overlap strategy with normalized embeddings, improving retrieval precision and reducing context noise for long-document Q&A.

Nexora — AI-Powered Startup Development Platform *Personal Project*
Python | FastAPI | Groq API | Multi-Agent LLM | Prompt Engineering GitHub

- Designed **multi-agent LLM orchestration** with 5 specialized agents (ideation, feasibility, competitor analysis, MVP scoping, pitch deck), cutting idea-to-plan cycle from hours to **under 2 minutes**.
- Built modular **FastAPI microservice backend** with prompt pipelines and response caching, reducing redundant LLM API calls by **∼60%** and improving throughput.

Education

Sri Venkateshwara College of Engineering, Bengaluru *Jul 2023 – Aug 2027*
B.E. in Computer Science — Artificial Intelligence **CGPA:** 7.8

Sheshadripuram PU College, Bengaluru *May 2021 – Mar 2023*
Class XII — Science (PCMC) **Percentage:** 82.5%

Achievements & Certifications

- **National Hackathon — VoisForTech, Pune (2024):** Ranked in **top 50 of 625+ teams** nationwide in AI/ML track. Certificate
- **Frontend Web Developer** — Infosys Springboard Certification. Certificate
- **Machine Learning Foundations** — Infosys Springboard Certification. Certificate

Additional Information

- **Languages:** English (Fluent), Hindi (Fluent), Kannada (Fluent)
- **Open Source:** All projects publicly available with documented READMEs and reproducible pipelines on GitHub